KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 22 · MATH

Ship Day in Duluth Harbor — Teacher Lesson Plan & Answer Key

Learning objectives: Students convert customary lengths (inches, feet, yards), weights (ounces, pounds) and capacities (cups, pints, quarts, gallons), including fractions of a unit and mixed measurements; convert metric units (millimeters, centimeters, meters, kilometers, grams, kilograms, milliliters, liters) by moving the decimal point; compare measurements by changing to the same unit; and solve multistep measurement problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Solveig (11), who lives in Duluth, Minnesota, on Lake Superior. Her Uncle Torvald is a deckhand on a harbor tugboat, Captain Teodoro pilots the tug, and Solveig spends the day on the dock and in the tug's galley. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Coordinate Plane Review (4 pts, 2 each, Week 21 skill). If a student misses both warm-up slides, replot one point on scrap paper before moving on.
3. Slide 2 — Warm-Up, Part 2: Distances and Patterns Review (4 pts, 2 each): a grid distance with a map scale and a corresponding term from two rules.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W22.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why a Harbor Runs on Measurement: nonfiction connection about the Port of Duluth-Superior, which moves more cargo by weight than any other Great Lakes port and is the farthest-inland freshwater seaport in North America (about 2,342 miles from the Atlantic). The Aerial Lift Bridge (built in 1905, made a lift bridge in 1929 and 1930) gives 135 feet of clearance and lifts about 5,000 times a year. No questions.
6. Slide 5 — Picture It: Yards, Feet, and Inches. 1 foot = 12 inches; 1 yard = 3 feet = 36 inches. Bigger unit to smaller unit, multiply; smaller to bigger, divide. 4 2/3 yards = 14 feet; 2 1/2 feet = 30 inches; 100 inches = 8 R4 = 8 feet 4 inches, a mixed measurement.
7. Slide 6 — Your Turn, Part 1: Yards, Feet, and Inches (8 pts, 4 each): fractional yards to feet and fractional feet or yards to inches. Watch for 3/4 foot read as 3 inches.
8. Slide 7 — Your Turn, Part 2: Mixed Measurements (4 pts): inches to feet and inches (divide by 12; the remainder is the inches). Common written forms, such as 9 ft 7 in, are accepted.
9. Slide 8 — Teach: Weight and Capacity in the Galley. 1 pound = 16 ounces; 1 gallon = 4 quarts = 8 pints = 16 cups. 3 gallons = 48 cups; 1/4 pound = 4 ounces; 56 ounces = 3 pounds 8 ounces = 3 1/2 pounds.
10. Slide 9 — Your Turn, Part 1: Weight and Capacity (8 pts, 4 each): gallons to cups and mixed-number pounds to ounces.

11. Slide 10 — Your Turn, Part 2: Smaller Units to Bigger Units (4 pts): divide, then write the leftover as a fraction (7 pints = 3 1/2 quarts).
12. Slide 11 — Teach: Metric Units and the Decimal Point. Kilo- means 1,000, centi- one hundredth, milli- one thousandth. 2.4 km = 2,400 m; 375 cm = 3.75 m; 2,750 mL = 2.75 L; 82 mm = 8.2 cm. Check: a smaller unit gives a bigger number.
13. Slide 12 — Your Turn, Part 1: Metric Conversions (8 pts, 4 each): one bigger-to-smaller and one smaller-to-bigger conversion.
14. Slide 13 — Your Turn, Part 2: Metric Multistep (4 pts): change to the same unit first, then finish the problem.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (customary system, metric system, convert, mixed measurement, capacity, weight and mass, kilo-, milli-).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Comparing Units and Multistep Problems. Same unit first, then solve: 2 1/2 pounds = 40 ounces, so 42 ounces is heavier; a 50-yard line minus 8 pieces of 15 feet leaves 150 − 120 = 30 feet (10 yards); 1.2 kg + 450 g = 1,650 g. Multistep plan: circle the answer's unit, convert, one step at a time, label.
18. Slide 17 — Your Turn: Thinking With Units (6 pts, 3 each): a cut-and-subtract or use-and-subtract problem and a comparison in two units.
19. Slide 18 — Real World: On the Dock (4 pts): a different situation in each set (drinking water in gallons and quarts, crate mass in grams, or name tags from a 3-yard ribbon).
20. Slide 19 — Real World: Galley and Lakewalk (4 pts): meals from a 5-pound bag of rice, a Lakewalk distance in meters, or water bottles in liters.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Both serial-comma and no-serial-comma forms of a list are accepted where a list appears.
24. Slide 23 — Match-Up Round (10 pts): 5 measurements matched to the equal measurement in another unit (for example, 2 yards = 72 inches; 48 ounces = 3 pounds), 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Big Freighter Arrives (8 pts, 4 each): a different situation in each set (a 30-yard tow line cut into pieces, 3 gallons of chowder in 1 1/2-cup bowls, or 16 boxes of bolts on a pallet, in kilograms and grams).
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W22.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Ship Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Duluth Harbor Measurement Master!" with the congratulations message "You kept the whole harbor running in Duluth!" and a big confetti burst; below 85 shows "Good Work on the Harbor!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: COORDINATE PLANE REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Coordinate plane (Week 21): Solveig starts at the origin of a harbor grid and moves 6 units right, then 2 units up. What ordered pair names where she stops? Ordered pair:	**(6, 2)** (also accepted: 6, 2, x = 6, y = 2, x=6,y=2)	Start at the origin. Move 6 units right (x = 6), then 2 units up (y = 2). The ordered pair is (6, 2).
r2_s0	Set 1: Coordinate plane (Week 21): which point lies on the y-axis? [(0, 5) / (5, 0) / (5, 5) / (1, 5)]	**(0, 5)**	A point on the y-axis has an x-coordinate of 0, because you move 0 units right. (5, 0) is on the x-axis.
r1_s1	Set 2: Coordinate plane (Week 21): Solveig starts at the origin of a harbor grid and moves 9 units right, then 4 units up. What ordered pair names where she stops? Ordered pair:	**(9, 4)** (also accepted: 9, 4, x = 9, y = 4, x=9,y=4)	Start at the origin. Move 9 units right (x = 9), then 4 units up (y = 4). The ordered pair is (9, 4).
r2_s1	Set 2: Coordinate plane (Week 21): which point lies on the x-axis? [(7, 0) / (0, 7) / (7, 7) / (7, 1)]	**(7, 0)**	A point on the x-axis has a y-coordinate of 0, because you move 0 units up. (0, 7) is on the y-axis.
r1_s2	Set 3: Coordinate plane (Week 21): Solveig starts at (2, 9) on a harbor grid. She moves 5 units right and then 3 units down. What ordered pair names where she stops? Ordered pair:	**(7, 6)** (also accepted: 7, 6, x = 7, y = 6, x=7,y=6)	Start at (2, 9). Moving right adds to x: 2 + 5 = 7. Moving down subtracts from y: 9 − 3 = 6. The ordered pair is (7, 6).
r2_s2	Set 3: Real-world graph (Week 21): a graph shows (hours, miles). Which point means “after 3 hours, 10 miles”? [(3, 10) / (10, 3) / (3, 13) / (0, 10)]	**(3, 10)**	The x-coordinate is the hours (3) and the y-coordinate is the miles (10), so the point is (3, 10).

WARM-UP: DISTANCES AND PATTERNS REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r3_s0	Set 1: Grid distance (Week 21): the dock is at (3, 8) and the bridge is at (3, 2). Each grid unit stands for 1/2 mile. How many miles apart are they along the grid line? Miles:	**3** (also accepted: 3 miles, 3 mile, 3 mi)	Both points have an x-coordinate of 3, so subtract the y-coordinates: 8 − 2 = 6 units. 6 × 1/2 = 3 miles.
r4_s0	Set 1: Two rules (Week 21): Rule A: start at 0, add 4. Rule B: start at 0, add 12. Which Rule B number corresponds to 28 in Rule A? Rule B number:	**84**	Rule B adds 12 and Rule A adds 4, so each Rule B number is 3 times the Rule A number. 28 × 3 = 84.
r3_s1	Set 2: Grid distance (Week 21): the dock is at (1, 5) and the grain elevator is at (9, 5). Each grid unit stands for 1/4 mile. How many miles apart are they along the grid line? Miles:	**2** (also accepted: 2 miles, 2 mile, 2 mi)	Both points have a y-coordinate of 5, so subtract the x-coordinates: 9 − 1 = 8 units. 8 × 1/4 = 2 miles.
r4_s1	Set 2: Two rules (Week 21): Rule A: start at 0, add 3. Rule B: start at 0, add 15. Which Rule B number corresponds to 27 in Rule A? Rule B number:	**135**	Rule B adds 15 and Rule A adds 3, so each Rule B number is 5 times the Rule A number. 27 × 5 = 135.
r3_s2	Set 3: Grid distance (Week 21): the canal is at (4, 2) and the boat ramp is at (4, 10). Each grid unit stands for 1/2 mile. How many miles apart are they along the grid line? Miles:	**4** (also accepted: 4 miles, 4 mile, 4 mi)	Both points have an x-coordinate of 4, so subtract the y-coordinates: 10 − 2 = 8 units. 8 × 1/2 = 4 miles.

ID	Question	Answer	Why (explain to student)
r4_s2	Set 3: Two rules (Week 21): Rule A: start at 0, add 6. Rule B: start at 0, add 2. Which Rule B number corresponds to 42 in Rule A? Rule B number:	**14**	Rule B adds 2 and Rule A adds 6, so each Rule B number is 1/3 of the Rule A number. 42 ÷ 3 = 14.

YOUR TURN: YARDS, FEET, AND INCHES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Uncle Torvald's mooring line is 4 2/3 yards long. How many feet long is it? Feet:	**14** (also accepted: 14 feet, 14 foot, 14 ft)	1 yard = 3 feet. 4 yards = 12 feet, and 2/3 of a yard = 2/3 × 3 = 2 feet. 12 + 2 = 14 feet.
l2_s0	Set 1: A spare rope on the tugboat is 2 3/4 feet long. How many inches long is it? [33 inches / 27 inches / 30 inches / 39 inches]	**33 inches**	2 feet = 24 inches, and 3/4 of a foot = 3/4 × 12 = 9 inches. 24 + 9 = 33 inches. (27 comes from 2 × 12 + 3, treating 3/4 as 3 inches.)
l1_s1	Set 2: Uncle Torvald's mooring line is 5 1/3 yards long. How many feet long is it? Feet:	**16** (also accepted: 16 feet, 16 foot, 16 ft)	1 yard = 3 feet. 5 yards = 15 feet, and 1/3 of a yard = 1/3 × 3 = 1 foot. 15 + 1 = 16 feet.
l2_s1	Set 2: A spare rope on the tugboat is 3 1/4 feet long. How many inches long is it? [39 inches / 40 inches / 36 inches / 42 inches]	**39 inches**	3 feet = 36 inches, and 1/4 of a foot = 1/4 × 12 = 3 inches. 36 + 3 = 39 inches. (40 comes from 3 × 12 + 4, treating 1/4 as 4 inches.)
l1_s2	Set 3: Uncle Torvald's mooring line is 6 2/3 yards long. How many feet long is it? Feet:	**20** (also accepted: 20 feet, 20 foot, 20 ft)	1 yard = 3 feet. 6 yards = 18 feet, and 2/3 of a yard = 2/3 × 3 = 2 feet. 18 + 2 = 20 feet.
l2_s2	Set 3: A canvas cover for a lifeboat is 1 1/2 yards long. How many inches long is it? [54 inches / 48 inches / 42 inches / 18 inches]	**54 inches**	1 yard = 36 inches, and 1/2 yard = 18 inches. 36 + 18 = 54 inches. (42 comes from 36 + 6, using half of a foot instead of half of a yard.)

YOUR TURN: MIXED MEASUREMENTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l3_s0	Set 1: Two steps: Solveig measures a hatch cover on the dock. It is 115 inches wide. Write the width in feet and inches. Feet and inches:	**9 feet 7 inches** (also accepted: 9 ft 7 in, 9 feet, 7 inches, 9 feet and 7 inches, 9 ft, 7 in, 9 ft. 7 in, 9' 7", 9 foot 7 inches, 9 feet 7 inch)	115 ÷ 12 = 9 R7, because 9 × 12 = 108 and 115 − 108 = 7. So 115 inches = 9 feet 7 inches.
l3_s1	Set 2: Two steps: Solveig measures a gangway railing. It is 87 inches long. Write the length in feet and inches. Feet and inches:	**7 feet 3 inches** (also accepted: 7 ft 3 in, 7 feet, 3 inches, 7 feet and 3 inches, 7 ft, 3 in, 7 ft. 3 in, 7' 3", 7 foot 3 inches, 7 feet 3 inch)	87 ÷ 12 = 7 R3, because 7 × 12 = 84 and 87 − 84 = 3. So 87 inches = 7 feet 3 inches.
l3_s2	Set 3: Two steps: Solveig measures a wooden dock bumper. It is 130 inches long. Write the length in feet and inches. Feet and inches:	**10 feet 10 inches** (also accepted: 10 ft 10 in, 10 feet, 10 inches, 10 feet and 10 inches, 10 ft, 10 in, 10 ft. 10 in, 10' 10", 10 foot 10 inches, 10 feet 10 inch)	130 ÷ 12 = 10 R10, because 10 × 12 = 120 and 130 − 120 = 10. So 130 inches = 10 feet 10 inches.

YOUR TURN: WEIGHT AND CAPACITY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Two steps: the galley makes 5 gallons of vegetable soup for the crew. How many cups of soup is that? Cups:	**80** (also accepted: 80 cups, 80 cup, 80 c)	1 gallon = 4 quarts = 16 cups. 5 gallons × 16 = 80 cups. (Or: 5 × 4 = 20 quarts, 20 × 2 = 40 pints, 40 × 2 = 80 cups.)
k2_s0	Set 1: A box of bolts weighs 3 3/4 pounds. How many ounces does it weigh? [60 ounces / 51 ounces / 64 ounces / 48 ounces]	**60 ounces**	3 pounds = 48 ounces, and 3/4 of a pound = 3/4 × 16 = 12 ounces. 48 + 12 = 60 ounces. (51 comes from 48 + 3, treating 3/4 as 3 ounces.)
k1_s1	Set 2: Two steps: the galley makes 7 gallons of chili for two crews. How many cups of chili is that? Cups:	**112** (also accepted: 112 cups, 112 cup, 112 c)	1 gallon = 16 cups. 7 gallons × 16 = 112 cups.
k2_s1	Set 2: A bag of washers weighs 2 1/4 pounds. How many ounces does it weigh? [36 ounces / 20 ounces / 34 ounces / 48 ounces]	**36 ounces**	2 pounds = 32 ounces, and 1/4 of a pound = 1/4 × 16 = 4 ounces. 32 + 4 = 36 ounces.
k1_s2	Set 3: Two steps: the galley makes 4 1/2 gallons of lemonade for a dock party. How many cups is that? Cups:	**72** (also accepted: 72 cups, 72 cup, 72 c)	4 gallons = 4 × 16 = 64 cups, and 1/2 gallon = 8 cups. 64 + 8 = 72 cups.
k2_s2	Set 3: A coil of steel wire weighs 5 1/2 pounds. How many ounces does it weigh? [88 ounces / 82 ounces / 80 ounces / 96 ounces]	**88 ounces**	5 pounds = 80 ounces, and 1/2 pound = 8 ounces. 80 + 8 = 88 ounces. (82 comes from 80 + 2.)

YOUR TURN: SMALLER UNITS TO BIGGER UNITS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k3_s0	Set 1: Two steps: the cook has 7 pints of soup. How many quarts is that? Write a mixed number. Quarts:	**3 1/2** (also accepted, with or without the unit word: 7/2, 3 and 1/2, 3.5)	2 pints make 1 quart, so divide: 7 ÷ 2 = 3 R1. The leftover pint is half of a quart, so 7 pints = 3 1/2 quarts.
k3_s1	Set 2: Two steps: the cook makes 10 cups of hot cocoa. How many quarts is that? Write a mixed number. Quarts:	**2 1/2** (also accepted, with or without the unit word: 5/2, 2 and 1/2, 2.5)	2 cups make 1 pint: 10 ÷ 2 = 5 pints. 2 pints make 1 quart: 5 ÷ 2 = 2 1/2 quarts. (Or: 4 cups make 1 quart, and 10 ÷ 4 = 2 1/2.)
k3_s2	Set 3: Two steps: a bag of sugar in the galley weighs 52 ounces. How many pounds is that? Write a mixed number. Pounds:	**3 1/4** (also accepted, with or without the unit word: 13/4, 3 and 1/4, 3.25)	16 ounces make 1 pound: 52 ÷ 16 = 3 R4. 4 ounces is 4/16 = 1/4 of a pound, so 52 ounces = 3 1/4 pounds.

YOUR TURN: METRIC CONVERSIONS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Captain Teodoro guides a ship 3.6 kilometers across the harbor. How many meters is that? Meters:	**3,600** (also accepted: 3,600 meters, 3,600 meter, 3,600 m)	1 kilometer = 1,000 meters. 3.6 × 1,000 = 3,600 meters. The decimal point moves 3 places to the right.
e2_s0	Set 1: A coil of cable is 465 centimeters long. How many meters long is it? [4.65 meters / 46.5 meters / 0.465 meter / 4,650 meters]	**4.65 meters**	100 centimeters = 1 meter, so divide by 100: 465 ÷ 100 = 4.65 meters. The decimal point moves 2 places to the left. (46.5 divides by 10, and 4,650 multiplies.)
e1_s1	Set 2: Captain Teodoro guides a ship 5.2 kilometers along the shore. How many meters is that? Meters:	**5,200** (also accepted: 5,200 meters, 5,200 meter, 5,200 m)	1 kilometer = 1,000 meters. 5.2 × 1,000 = 5,200 meters. The decimal point moves 3 places to the right.

ID	Question	Answer	Why (explain to student)
e2_s1	Set 2: A jug of cleaning water holds 1,850 milliliters. How many liters is that? [1.85 liters / 18.5 liters / 0.185 liter / 185 liters]	**1.85 liters**	1,000 milliliters = 1 liter, so divide by 1,000: 1,850 ÷ 1,000 = 1.85 liters. The decimal point moves 3 places to the left.
e1_s2	Set 3: A life ring rope is 2.35 meters long. How many centimeters long is it? Centimeters:	**235** (also accepted: 235 centimeters, 235 centimeter, 235 cm)	1 meter = 100 centimeters. 2.35 × 100 = 235 centimeters. The decimal point moves 2 places to the right.
e2_s2	Set 3: A crate of engine parts has a mass of 6,400 grams. What is its mass in kilograms? [6.4 kilograms / 64 kilograms / 0.64 kilogram / 640 kilograms]	**6.4 kilograms**	1,000 grams = 1 kilogram, so divide by 1,000: 6,400 ÷ 1,000 = 6.4 kilograms. The decimal point moves 3 places to the left.

YOUR TURN: METRIC MULTISTEP (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e3_s0	Set 1: Two steps: Uncle Torvald loads 4 boxes of shackles. Each box has a mass of 0.75 kilogram. What is the total mass in grams? Grams:	**3,000** (also accepted: 3,000 grams, 3,000 gram, 3,000 g)	One box: 0.75 kilogram × 1,000 = 750 grams. Four boxes: 750 × 4 = 3,000 grams. (Or: 0.75 × 4 = 3 kilograms = 3,000 grams.)
e3_s1	Set 2: Two steps: each bolt is 38 millimeters long. Solveig lines up 10 bolts end to end. How many centimeters long is the line? Centimeters:	**38** (also accepted: 38 centimeters, 38 centimeter, 38 cm)	Ten bolts: 38 × 10 = 380 millimeters. 10 millimeters = 1 centimeter: 380 ÷ 10 = 38 centimeters.
e3_s2	Set 3: Three steps: the cook pours 4 full pitchers of juice, each holding 1.5 liters, into cups that each hold 250 milliliters. How many cups can she fill? Cups:	**24** (also accepted: 24 cups, 24 cup)	Four pitchers: 1.5 × 4 = 6 liters = 6,000 milliliters. Each cup holds 250 milliliters: 6,000 ÷ 250 = 24 cups.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: “The prefix kilo- means 1,000.” How many meters are in 1 kilometer? [1,000 meters / 100 meters / 10 meters / 1 meter]	**1,000 meters**	Kilo- means 1,000, so 1 kilometer = 1,000 meters, and 1 kilogram = 1,000 grams.
v2_s0	Set 1: Type your answer: the prefix ___- means one thousandth, as in millimeter and milliliter.	**milli** (also accepted: milli-)	Milli- means one thousandth: 1 millimeter is 1/1,000 of a meter, and 1 milliliter is 1/1,000 of a liter.
v1_s1	Set 2: From the lesson: “Cups, pints, quarts, and gallons measure capacity.” Which pair of units BOTH measure capacity? [Quarts and liters / Pounds and grams / Feet and meters / Inches and kilograms]	**Quarts and liters**	Capacity is the amount a container can hold. Quarts (customary) and liters (metric) both measure capacity. Pounds and grams measure weight and mass.
v2_s1	Set 2: Type your answer: to change a bigger unit into a smaller unit, you ___. (one word)	**multiply** (also accepted: multiplication, times, multiplying)	A bigger unit holds many smaller units, so you multiply: 3 yards × 36 = 108 inches.
v1_s2	Set 3: From the lesson: “A measurement that uses two units at once, like 8 feet 4 inches, is a mixed measurement.” Which is a mixed measurement? [3 pounds 8 ounces / 56 ounces / 3.5 pounds / 3 1/2 pounds]	**3 pounds 8 ounces**	A mixed measurement uses two units at once. 3 pounds 8 ounces uses pounds and ounces. The other choices each use only one unit, even though they are equal amounts.

ID	Question	Answer	Why (explain to student)
v2_s2	Set 3: Type your answer: to change a smaller unit into a bigger unit, you ___. (one word)	**divide** (also accepted: division, dividing)	It takes many smaller units to make one bigger unit, so you divide: 48 ounces ÷ 16 = 3 pounds.

YOUR TURN: THINKING WITH UNITS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Three steps: Uncle Torvald has a 25-yard tow line. He cuts 4 pieces that are each 12 feet long. How many feet of line are left? Feet:	**27** (also accepted: 27 feet, 27 foot, 27 ft)	The line is 25 × 3 = 75 feet. The pieces use 4 × 12 = 48 feet. 75 − 48 = 27 feet left.
t2_s0	Set 1: Which is longer: 2 1/4 yards or 80 inches? [2 1/4 yards, because it equals 81 inches / 80 inches, because 80 is greater than 2 1/4 / They are the same length / 80 inches, because it equals 2 1/2 yards]	**2 1/4 yards, because it equals 81 inches**	2 1/4 yards = 2 × 36 + 1/4 × 36 = 72 + 9 = 81 inches. 81 inches is 1 inch longer than 80 inches. You cannot compare 80 and 2 1/4 until the units match.
t1_s1	Set 2: Two steps: the galley has 3 gallons of milk. The cook uses 5 quarts for pancakes. How many quarts of milk are left? Quarts:	**7** (also accepted: 7 quarts, 7 quart, 7 qt)	3 gallons = 3 × 4 = 12 quarts. 12 − 5 = 7 quarts left.
t2_s1	Set 2: Which is heavier: 3 1/2 pounds or 54 ounces? [3 1/2 pounds, because it equals 56 ounces / 54 ounces, because 54 is greater than 3 1/2 / They weigh the same / 54 ounces, because it equals 4 pounds]	**3 1/2 pounds, because it equals 56 ounces**	3 1/2 pounds = 3 × 16 + 8 = 56 ounces. 56 ounces is 2 ounces more than 54 ounces. (54 ounces = 3 pounds 6 ounces.)
t1_s2	Set 3: Two steps: a spool holds 4.5 meters of cable. Solveig cuts off 175 centimeters. How many centimeters are left? Centimeters:	**275** (also accepted: 275 centimeters, 275 centimeter, 275 cm)	The spool holds 4.5 × 100 = 450 centimeters. 450 − 175 = 275 centimeters left.
t2_s2	Set 3: Which holds more: a 1.2-liter bottle or a 1,150-milliliter bottle? [The 1.2-liter bottle, because it holds 1,200 milliliters / The 1,150-milliliter bottle, because 1,150 is greater than 1.2 / They hold the same amount / The 1,150-milliliter bottle, because it holds 11.5 liters]	**The 1.2-liter bottle, because it holds 1,200 milliliters**	1.2 liters × 1,000 = 1,200 milliliters. 1,200 milliliters is 50 milliliters more than 1,150 milliliters.

REAL WORLD: ON THE DOCK (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Uncle Torvald's tugboat carries 9 gallons of drinking water for a long day of work. Each of the 6 crew members drinks 2 quarts during the day.* Three steps: how many gallons of water are left at the end of the day? Gallons:	**6** (also accepted: 6 gallons, 6 gallon, 6 gal)	The tug starts with 9 × 4 = 36 quarts. The crew drinks 6 × 2 = 12 quarts. 36 − 12 = 24 quarts left, and 24 ÷ 4 = 6 gallons.
w1_s1	Set 2: *The Situation A cart on the dock holds 15 crates of machine parts. Each crate has a mass of 2.4 kilograms.* Two steps: what is the total mass of the 15 crates in grams? [36,000 grams / 3,600 grams / 360 grams / 36 grams]	**36,000 grams**	15 × 2.4 = 36 kilograms. 36 × 1,000 = 36,000 grams. (36 is the answer in kilograms, not grams.)

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: *The Situation Solveig makes name tags for a harbor tour from a ribbon that is 3 yards long. Each name tag uses 8 inches of ribbon.* Two steps: how many full name tags can Solveig make? Name tags:	**13** (also accepted with a unit word, such as 13 tags or 13 full tags, or 2 other unit spellings)	The ribbon is 3 × 36 = 108 inches. 108 ÷ 8 = 13 R4, so she can make 13 full name tags, with 4 inches left over.

REAL WORLD: GALLEY AND LAKEWALK (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation The galley cook has a 5-pound bag of rice. Each crew meal uses 10 ounces of rice.* Two steps: how many full meals can the cook make with the bag of rice? [8 meals / 50 meals / 2 meals / 6 meals]	**8 meals**	5 pounds = 5 × 16 = 80 ounces. 80 ÷ 10 = 8 meals. (50 comes from 5 × 10, and 2 comes from 10 ÷ 5.)
w2_s1	Set 2: *The Situation After work, Solveig walks on Duluth's Lakewalk, a path along the shore of Lake Superior. She walks 2.5 kilometers in the morning and 1,800 meters in the evening.* Two steps: how many meters does Solveig walk in all? Meters:	**4,300** (also accepted: 4,300 meters, 4,300 meter, 4,300 m)	Change to meters: 2.5 kilometers = 2,500 meters. 2,500 + 1,800 = 4,300 meters.
w2_s2	Set 3: *The Situation Before a harbor tour, Solveig fills 12 water bottles. Each bottle holds 750 milliliters.* Two steps: how many liters of water are in all 12 bottles? [9 liters / 90 liters / 0.9 liter / 9,000 liters]	**9 liters**	12 × 750 = 9,000 milliliters. 9,000 ÷ 1,000 = 9 liters. (9,000 is the answer in milliliters.)

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? How many inches are in 3/4 yard? [27 inches, because 1 yard = 36 inches and 3/4 of 36 is 27 / 9 inches, because 3/4 of 12 is 9 / 75 inches, because 3/4 = 0.75 / 4 inches, because 36 ÷ 9 = 4]	**27 inches, because 1 yard = 36 inches and 3/4 of 36 is 27**	A yard is 36 inches, not 12. 36 ÷ 4 = 9, so 1/4 yard = 9 inches and 3/4 yard = 27 inches. 9 inches is 3/4 of a foot.
x2_s0	Set 1: Find the mistake: Stellan says 4.3 kilometers = 430 meters. What did he do wrong? [He multiplied by 100. 1 kilometer = 1,000 meters, so 4.3 × 1,000 = 4,300 meters / He should divide: 4.3 ÷ 1,000 = 0.0043 meter / He should multiply by 10: 43 meters / Nothing. 430 meters is correct]	**He multiplied by 100. 1 kilometer = 1,000 meters, so 4.3 × 1,000 = 4,300 meters**	Kilo- means 1,000. Changing kilometers to meters means multiplying by 1,000: the decimal point moves 3 places to the right, so 4.3 km = 4,300 m.
x1_s1	Set 2: Which explanation is correct? How many quarts are in 20 cups? [5 quarts, because 4 cups make 1 quart and 20 ÷ 4 = 5 / 80 quarts, because 20 × 4 = 80 / 10 quarts, because 2 cups make 1 quart / 2 1/2 quarts, because 20 ÷ 8 = 2 1/2]	**5 quarts, because 4 cups make 1 quart and 20 ÷ 4 = 5**	1 quart = 2 pints = 4 cups. Cups are smaller, so divide: 20 ÷ 4 = 5 quarts. Two cups make a pint, not a quart.

ID	Question	Answer	Why (explain to student)
x2_s1	Set 2: Find the mistake: Tamsin says 3 pounds 5 ounces = 35 ounces. What should she fix? [Multiply the pounds by 16, then add: 3 × 16 + 5 = 53 ounces / Multiply by 10: 3 × 10 + 5 = 35 ounces / Divide: 35 ÷ 16 ounces / Nothing. 35 ounces is correct]	**Multiply the pounds by 16, then add: 3 × 16 + 5 = 53 ounces**	Tamsin just wrote the digits 3 and 5 side by side. Each pound is 16 ounces: 3 × 16 = 48, and 48 + 5 = 53 ounces.
x1_s2	Set 3: Which explanation is correct? Which is longer, 1.6 meters or 155 centimeters? [1.6 meters, because it equals 160 centimeters / 155 centimeters, because 155 is greater than 1.6 / They are the same length / 155 centimeters, because it equals 15.5 meters]	**1.6 meters, because it equals 160 centimeters**	1.6 × 100 = 160 centimeters. 160 centimeters is 5 centimeters longer than 155 centimeters. Change to the same unit before comparing.
x2_s2	Set 3: Find the mistake: Soren says 74 inches is 7 feet 4 inches. What should he fix? [Divide by 12: 74 ÷ 12 = 6 R2, so it is 6 feet 2 inches / Nothing. 7 feet 4 inches is correct / Multiply by 12: 74 × 12 = 888 feet / Divide by 3: 74 ÷ 3 = 24 feet 2 inches]	**Divide by 12: 74 ÷ 12 = 6 R2, so it is 6 feet 2 inches**	There are 12 inches in a foot, not 10. 6 × 12 = 72, and 74 − 72 = 2, so 74 inches = 6 feet 2 inches. Check: 7 feet 4 inches would be 88 inches.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: solveig asked, how many feet are in a yard	**Solveig asked, "How many feet are in a yard?"**	Capitalize the name Solveig, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The tugboat turned it's bow toward the canal.	**The tugboat turned its bow toward the canal.**	Its (no apostrophe) shows ownership: the bow belongs to the tugboat. It's always means it is.
s3_s0	Set 1: Fix this sentence: The ship carried grain iron ore and coal.	**The ship carried grain, iron ore, and coal.** (also accepted: The ship carried grain, iron ore and coal.)	Use commas to separate the items in a series: grain, iron ore, and coal.
s4_s0	Set 1: Fix this sentence: the ships in the harbor waits for the bridge to rise	**The ships in the harbor wait for the bridge to rise.** (also accepted: The ships in the harbor waited for the bridge to rise.)	Capitalize the first word. The subject is ships (plural), not harbor, so use wait, not waits. End with a period.
s1_s1	Set 2: Fix this sentence: uncle torvald said, tie the line to the cleat	**Uncle Torvald said, "Tie the line to the cleat."**	Capitalize Uncle Torvald (a title used with a name), put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The sailors hung there jackets near the galley.	**The sailors hung their jackets near the galley.**	Their shows ownership: the jackets belong to the sailors. There tells where something is.
s3_s1	Set 2: Fix this sentence: When the bridge went up the crowd cheered.	**When the bridge went up, the crowd cheered.**	Put a comma after an opening clause that begins with when: When the bridge went up, the crowd cheered.

ID	Question	Answer	Why (explain to student)
s4_s1	Set 2: Fix this sentence: captain teodoro measure the rope in meters	**Captain Teodoro measures the rope in meters.** (also accepted: Captain Teodoro measured the rope in meters.)	Capitalize Captain Teodoro (a title used with a name). One person needs the verb measures, not measure. End with a period.
s1_s2	Set 3: Fix this sentence: solveig said, a liter is 1,000 milliliters	**Solveig said, "A liter is 1,000 milliliters."**	Capitalize Solveig, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Its cold on Lake Superior in November.	**It's cold on Lake Superior in November.**	It's means it is: it is cold. Use the apostrophe.
s3_s2	Set 3: Fix this sentence: The cook measured cups pints and quarts.	**The cook measured cups, pints, and quarts.** (also accepted: The cook measured cups, pints and quarts.)	Use commas to separate the items in a series: cups, pints, and quarts.
s4_s2	Set 3: Fix this sentence: the crates on the dock weighs 40 pounds each	**The crates on the dock weigh 40 pounds each.** (also accepted: The crates on the dock weighed 40 pounds each.)	Capitalize the first word. The subject is crates (plural), not dock, so use weigh, not weighs. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 2 yards	**72 inches**	2 points if matched on the first try.
match_s0_2	Set 1: Match 3 gallons	**12 quarts**	2 points if matched on the first try.
match_s0_3	Set 1: Match 2.5 kilograms	**2,500 grams**	2 points if matched on the first try.
match_s0_4	Set 1: Match 48 ounces	**3 pounds**	2 points if matched on the first try.
match_s0_5	Set 1: Match 350 centimeters	**3.5 meters**	2 points if matched on the first try.
match_s1_1	Set 2: Match 4 feet	**48 inches**	2 points if matched on the first try.
match_s1_2	Set 2: Match 5 pints	**10 cups**	2 points if matched on the first try.
match_s1_3	Set 2: Match 1.75 liters	**1,750 milliliters**	2 points if matched on the first try.
match_s1_4	Set 2: Match 40 ounces	**2 1/2 pounds**	2 points if matched on the first try.
match_s1_5	Set 2: Match 6.2 kilometers	**6,200 meters**	2 points if matched on the first try.
match_s2_1	Set 3: Match 2 1/3 yards	**7 feet**	2 points if matched on the first try.
match_s2_2	Set 3: Match 2 quarts	**8 cups**	2 points if matched on the first try.
match_s2_3	Set 3: Match 0.8 kilogram	**800 grams**	2 points if matched on the first try.
match_s2_4	Set 3: Match 24 ounces	**1 1/2 pounds**	2 points if matched on the first try.
match_s2_5	Set 3: Match 90 millimeters	**9 centimeters**	2 points if matched on the first try.

BOSS BATTLE: THE BIG FREIGHTER ARRIVES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Solveig and Uncle Torvald Ready the Lines Uncle Torvald has a tow line that is 30 yards long. He cuts off a 14-foot piece to repair a bumper. Then he cuts the rest of the line into 4 pieces of equal length.* Two steps: how many feet of line are left after the first cut? [76 feet / 16 feet / 86 feet / 26 feet]	**76 feet**	30 yards × 3 = 90 feet. 90 − 14 = 76 feet. (16 comes from subtracting 14 from 30 without changing yards to feet.)
b2_s0	Set 1: Two steps: how long is each of the 4 equal pieces, in feet? Feet:	**19** (also accepted: 19 feet, 19 foot, 19 ft)	After the first cut, 76 feet are left. 76 ÷ 4 = 19 feet for each piece. Check: 4 × 19 = 76.
b1_s1	Set 2: *Solveig and the Galley Cook Feed the Crew The galley cook makes 3 gallons of fish chowder. Each crew bowl holds 1 1/2 cups of chowder.* How many cups of chowder are in 3 gallons? [48 cups / 12 cups / 24 cups / 96 cups]	**48 cups**	1 gallon = 16 cups, so 3 × 16 = 48 cups. (12 is the number of quarts, and 24 is the number of pints.)
b2_s1	Set 2: Two steps: how many bowls can the cook fill? Bowls:	**32** (also accepted: 32 bowls, 32 bowl)	There are 48 cups. Two bowls use 1 1/2 + 1 1/2 = 3 cups, and 48 ÷ 3 = 16, so 16 × 2 = 32 bowls. (Or: 48 ÷ 1 1/2 = 32.)
b1_s2	Set 3: *Solveig and Captain Teodoro Move the Cargo A forklift on the dock carries 16 boxes of bolts for wind turbine parts. Each box has a mass of 1.25 kilograms. The wooden pallet under the boxes has a mass of 22 kilograms.* What is the total mass of the 16 boxes, in kilograms? [20 kilograms / 2 kilograms / 200 kilograms / 17.25 kilograms]	**20 kilograms**	16 × 1.25 = 20 kilograms. (4 boxes make 5 kilograms, and 16 boxes is 4 groups of 4, so 4 × 5 = 20.)
b2_s2	Set 3: Two steps: what is the total mass of the boxes AND the pallet, in grams? Grams:	**42,000** (also accepted: 42,000 grams, 42,000 gram, 42,000 g)	Boxes and pallet: 20 + 22 = 42 kilograms. 42 × 1,000 = 42,000 grams.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W22.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: a) 4 right, 6 up b) Rule A add 2, Rule B add 10: partner of 6	a) (4, 6) b) 30. Rule A: 0, 2, 4, 6; Rule B: 0, 10, 20, 30. Each Rule B number is 5 times the Rule A number: 6 × 5 = 30.
2	Fill in: foot, yard and pound facts	1 foot = 12 inches. 1 yard = 3 feet. 1 yard = 36 inches. 1 pound = 16 ounces.
3	A 3-yard rope in feet and in inches	3 × 3 = 9 feet; 3 × 36 = 108 inches (or 9 × 12 = 108).
4	Best unit for the mass of a crate of bolts	kilograms

#	Prompt	Model answer / grading note
5	Quarts in 2 1/2 gallons; draw it	10 quarts. 2 gallons = 8 quarts and 1/2 gallon = 2 quarts; 8 + 2 = 10. The picture shows two gallons of 4 quarts each and a half gallon of 2 quarts.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	a) 3 2/3 yd to feet b) 1 3/4 ft to inches c) 95 in to feet and inches	a) 9 + 2 = 11 feet b) 12 + 9 = 21 inches c) 95 ÷ 12 = 7 R11, so 7 feet 11 inches.
2	a) 2 3/4 lb to ounces b) 6 gal to cups c) 9 pt to quarts	a) 32 + 12 = 44 ounces b) 6 × 16 = 96 cups c) 9 ÷ 2 = 4 R1, so 4 1/2 quarts.
3	a) 4.8 km to m b) 560 cm to m c) 3,250 mL to L	a) 4.8 × 1,000 = 4,800 meters b) 560 ÷ 100 = 5.6 meters c) 3,250 ÷ 1,000 = 3.25 liters
4	Heavier: 1 3/4 pounds or 30 ounces?	30 ounces. 1 3/4 pounds = 16 + 12 = 28 ounces, and 30 ounces is 2 ounces more.
5	Sylvie says 2.6 liters = 26 milliliters	Model: Sylvie multiplied by 10, moving the decimal point only one place. 1 liter is 1,000 milliliters, so she should multiply by 1,000 and move the decimal point three places to the right. The correct answer is 2,600 milliliters.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	40-yard line; 6 pieces of 14 feet	40 yards = 120 feet. The pieces use 6 × 14 = 84 feet. 120 − 84 = 36 feet left, which is 36 ÷ 3 = 12 yards.